

CHONGXING FAN

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EDUCATION

University of Michigan, Ann Arbor (UM)

August 2019 – May 2024

Ph.D. in Climate and Space Sciences and Engineering

Certificate of Graduate Studies in Computational Discovery and Engineering (2022)

GPA: 4.00/4.00

Ph.D. Dissertation: The Importance of Accurate Physical Parameterization for Radiative Transfer in Global Climate Simulations and Remote Sensing: Examples of Cloud Longwave Scattering and Solar Farm Modeling.

Advisor: Prof. Xianglei Huang

Nanjing University (NJU)

September 2015 – July 2019

Bachelor of Science in Atmospheric Sciences

GPA: 3.95/4.00

EMPLOYMENT

Princeton University

June 2024 – Present

Postdoctoral Research Associate

Program in Atmospheric and Oceanic Sciences

RESEARCH INTERESTS

- Atmospheric Radiative Transfer and Energy Budget
- Climate Modeling
- Application of Machine Learning in Climate Sciences

HONORS AND AWARDS

ProQuest Distinguished Dissertation Award Honorable Mention	2024
AGU Outstanding Student Presentation Award	2024
Richard and Eleanor Towner Prize for Distinguished Academic Achievement Award (UM)	2023
Future Investigators in NASA Earth and Space Science and Technology Fellowship (\$150,000, NASA)	2022

Project Title: Impacts of Solar Farming on Surface Energy Budget and Climate from Long-Term NASA Satellite Observations

Rackham International Students Fellowship (UM)	2020
MICDE Fellowship (UM)	2019
Honorable Mention in the Mathematical Contest in Modeling (MCM, COMAP)	2018
Chow Tai Fook Scholarship (Top 1%, NJU)	2018
China's National Scholarship (Top 1%, NJU)	2017
Scholarship of Mr. Liao (Top 1%, NJU)	2016

PROFESSIONAL SERVICES

- **Peer reviewer** for journals
Advances in Atmospheric Sciences (2022; 2023), *Journal of Climate* (2023; 2025), *Journal of Geophysical Research – Atmospheres* (2023~2024), *Atmospheric Chemistry and Physics* (2023), *Environmental Research: Climate* (2024), *Journal of the Atmospheric Sciences* (2024), *Geoscientific Model Development* (2024~2025), *Environmental Research Communications* (2024), *Atmospheric Science Letters* (2024), *Journal of Quantitative Spectroscopy and Radiative Transfer* (2024), *Environmental Research Letters* (2024~2025), *Remote Sensing Letters* (2025), *Remote Sensing of Environment* (2025)
- Selected as **IOP Trusted Reviewer** in 2024
- **Outstanding Reviewer** for *Environmental Research: Climate* in **IOP Outstanding Reviewer Awards 2024**

PEER-REVIEWED PUBLICATIONS

1. **Fan, C.**, Paynter, D., Kramer, R., & Lin, P. (2025). Sensitivity of Earth's Radiation Budget in Historical Climate Simulations to Lower Boundary Condition Datasets. *Geophysical Research Letters*, 52, e2025GL115914. <https://doi.org/10.1029/2025GL115914>
2. **Fan, C.**, & Huang, X. (2025). Monte-Carlo-Based Simulations of radiative transfer in photovoltaic solar farms: towards fast and accurate parameterizations in weather and climate models. *Journal of Quantitative Spectroscopy and Radiative Transfer*, 109417. <https://doi.org/10.1016/j.jqsrt.2025.109417>
3. **Fan, C.**, & Huang, X. (2023). Infrared scattering of cloud in an isothermal atmosphere. *Journal of the Atmospheric Sciences*, 80(11), 2701-2710. <https://doi.org/10.1175/JAS-D-23-0050.1>
4. **Fan, C.**, Chen, Y.-H., Chen, X., Lin, W., Yang, P., & Huang, X. (2023). A refined understanding of the ice cloud longwave scattering effects in climate model. *Journal of*

Advances in Modeling Earth Systems, 15, e2023MS003810.

<https://doi.org/10.1029/2023MS003810>

5. Huang, X., Chen, X., **Fan, C.**, Kato, S., Loeb, N., Bosilovich, M., et al. (2022). A synopsis of AIRS global-mean clear-sky radiance trends from 2003 to 2020. *Journal of Geophysical Research: Atmospheres*, 127, e2022JD037598. <https://doi.org/10.1029/2022JD037598>
6. **Fan, C.**, & Huang, X. (2021). Direct impact of solar farm deployment on surface longwave radiation. *Environmental Research Communications*, 3(12), 125006. <https://doi.org/10.1088/2515-7620/ac40f1>
7. **Fan, C.**, & Huang, X. (2020). Satellite-observed changes of surface spectral reflectances due to solar farming and the implication for radiation budget. *Environmental Research Letters*, 15(11), 114047. <https://doi.org/10.1088/1748-9326/abbdea>
8. **Fan, C.**, Wang, M., Rosenfeld, D., Zhu, Y., Liu, J., & Chen, B. (2020). Strong precipitation suppression by aerosols in marine low clouds. *Geophysical Research Letters*, 47(7), e2019GL086207. <https://doi.org/10.1029/2019GL086207>

CONFERENCES, PROCEEDINGS, AND ABSTRACTS

[O] Oral Talks; [P] Posters

1. [P] **Fan, C.**, Paynter, D., Kramer, R., & Lin, P. Assessing TOA Radiation Budget in GFDL AM4 with Different Prescribed Sea Surface Temperature and Sea Ice Datasets: Comparison to Satellite Products. GRC Radiation and Climate. Lewiston, ME, USA. July 20-25, 2025.
2. [O] **Fan, C.**, Paynter, D., Kramer, R., & Lin, P. Assessing Radiation Biases in GFDL AM4 due to Prescribed SST/SIC Datasets: Comparison to CERES Products. Gordon Research Seminar (GRS) Radiation and Climate. Lewiston, ME, USA. July 19-20, 2025.
3. [O] **Fan, C.**, Paynter, D., Kramer, R., & Lin, P. Assessing Radiation Biases in GFDL AM4 due to Prescribed SST/SIC Datasets: Comparison to CERES Products. Spring 2025 CERES Science Meeting. Hampton, VA, USA. May 13-15, 2025.
4. [P] **Fan, C.**, Paynter, D., & Kramer, R. Long-Term Comparison of 20-Year Spectral OLR Trends between the GFDL AM4 Model and Satellite Observations with Insights on Climate Model Biases. AGU Annual Meeting 2024. Washington D.C., USA. December 9-13, 2024.
5. [P] **Fan, C.**, & Huang, X. A Fast and Physically Accurate Radiation Parameterization for Large-Scale Solar Farms. AGU Annual Meeting 2024. Washington D.C., USA. December 9-13, 2024.
6. [P] **Fan, C.**, Chen, Y.-H., Chen, X., Lin, W., Yang, P., & Huang, X. Improving the Surface and Cloud Longwave Radiative Couplings in the Earth System Model: A Tribute to Prof. Liou's Works on Ice Cloud Scattering and Radiation-Topography Interaction. Kuo-Nan Liou Symposium, 104th AMS Annual Meeting. Baltimore, MD, USA. January 28-February

- 1, 2024.
7. **[O] Fan, C.,** Chen, Y.-H., Chen, X., Lin, W., Huang, X, & Yang, P. Address Climate Model Bias by Refining Radiation Scheme: Examples and Future Perspectives. AGU Fall Meeting 2023. San Francisco, CA, USA. December 11-15, 2023.
 8. **[O] Huang, X.,** Chen, X., Strow, L., **Fan, C.,** Loeb, N., Kato, S., Yue, Q. The Insights from Twenty Years of AIRS Radiances and an Outlook for the Incoming Decade: A Climate Perspective. 20th Annual Meeting of the Asia Oceania Geosciences Society. Singapore. July 30 - August 4, 2023.
 9. **[P] Fan, C., & Huang, X.** Understanding How Solar Farms Modify Radiation Budget and Regional Climate Based on Satellite-observation Constrained Climate Modeling. 20th Annual Meeting of the Asia Oceania Geosciences Society. Singapore. July 30 - August 4, 2023.
 10. **[O] Fan, C.,** Chen, Y.-H., Chen, X., Lin, W., Huang, X, & Yang, P. Including Ice-cloud Longwave Scattering and Surface Spectral Emissivities in Climate Models Leads to More Impacts on Mean-state Climate Than Climate Feedbacks. 20th Annual Meeting of the Asia Oceania Geosciences Society. Singapore. July 30 - August 4, 2023.
 11. **[P] Fan, C.,** Chen, Y.-H., Chen, X., Lin, W., Huang, X, & Yang, P. Ice-Cloud Longwave Scattering in Climate Models Leads to More Impacts on Mean-State Climate than Climate Feedbacks. GRC Radiation and Climate. Lewiston, Maine, USA. July 23-28, 2023.
 12. **[O] Fan, C.,** Chen, Y.-H., Chen, X., Lin, W., Huang, X., Yang, P., Including two missing physics for a more realistic representation of cloud-surface longwave radiative coupling in the earth system model. XXVIII General Assembly of the International Union of Geodesy and Geophysics (IUGG). Berlin, Germany. July 11-20, 2023.
 13. **[O] Fan, C.,** Chen, Y.-H., Chen, X., Lin, W., Huang, X, & Yang, P., Including Ice-Cloud Longwave Scattering and Surface Spectral Emissivities in Climate Models Leads to More Impacts on Mean-State Climate than Climate Feedbacks. EGU General Assembly 2023. Vienna, Austria. April 23-28, 2023.
 14. **[O] Fan, C.,** Chen, Y.-H., Jing, X., Chen, X., Lin, W., Huang, X, & Yang, P., An Overall Assessment of the Ice-Cloud Longwave Scattering Effects on the Simulated Global Climate. 36th Conference on Climate Variability and Change, 103rd AMS Annual Meeting. Denver, CO, USA. January 8-12, 2023.
 15. **[O] Fan, C., & Huang, X.,** Satellite-observed changes of surface reflectance, emissivity, and temperature due to solar farming and the implication for radiation budget. 14th Conference on Weather, Climate, and the New Energy Economy, 103rd AMS Annual Meeting. Denver, CO, USA. January 8-12, 2023.
 16. **[O] Fan, C.,** Chen, Y.-H., Jing, X., Chen, X., Lin, W., Huang, X, & Yang, P., An Overall

- Assessment of the Ice-Cloud Longwave Scattering Effects on the Simulated Global Climate. AGU Fall Meeting 2022. Chicago, IL, USA. December 12-16, 2022.
17. **[P]** Fan, C., & Huang, X., Satellite-Observed Changes of Surface Radiative Properties due to Solar Farming and the Implication for Radiation Budget. Midwest Student Conference on Atmospheric Research 2022. Urbana, IL, USA. October 1-2, 2022.
 18. **[P]** Fan, C., Chen, Y.-H., Jing, X., Chen, X., Lin, W., Huang, X., & Yang, P., Cloud scattering and surface spectral emissivities in climate model: Performance evaluation and feedback analysis. 2022 CFMIP Meeting on Clouds, Precipitation, Circulation and Climate Sensitivity. Seattle, WA, USA. July 19-22, 2022
 19. **[O]** Huang, X., Yang, P., Chen, Y.-H., Chen, X., Lin, W., Fan, C., High-latitude surface-atmosphere radiative coupling in the far-IR: missing physics in climate models and opportunities in future observations. 2022 International Radiation Symposium. Thessaloniki, Greece. July 4-8, 2022.
 20. **[O]** Fan, C., & Huang, X., Satellite-Observed Changes of Surface Radiative Properties due to Solar Farming and the Implication for Radiation Budget. 2022 International Radiation Symposium. Thessaloniki, Greece. July 4-8, 2022.
 21. **[P]** Fan, C., & Huang, X., Solar Farm as an ideal test bed for satellite surface emissivity and temperature retrieval algorithms. AGU Fall Meeting 2021. New Orleans, LA, USA. December 13-17, 2021.
 22. **[O]** Fan, C., & Huang, X., Use different machine-learning algorithms for clear-sky detections in infrared hyperspectral observations: assessment and physical interpretability. 3rd NOAA Workshop on Leveraging AI in Environmental Sciences. Online. September 13-17, 2021.
 23. **[P]** Fan, C., & Huang, X., Satellite-observed changes of surface spectral reflectances due to solar farming and the implication for radiation budget. AGU Fall Meeting 2020. Online. December 1-17, 2020.

INTERNSHIP EXPERIENCE

Globalink Research Internship

July 2018 – October 2018

- Project Title: Evaluation of quantitative precipitation estimation from model, satellite and radar
- Advisor: Prof. Yongsheng Chen (York University, Canada)

Meteorological Bureau of Hunan Province, China

February 2018

- Weather forecast intern

TEACHING AND MENTORING EXPERIENCE

Grader for CLIMATE 586 (Advanced Data Analysis) *September - December 2023*

- Responsibility: grading assignments; hosting office hours to answer students' questions

Grader for CLIMATE 586 (Advanced Data Analysis) *September - December 2022*

- Responsibility: grading assignments

Freshman Peer Advisor *November 2017 - June 2018*

- Responsibility: offering academic, career planning, and life advice

COMMUNITY SERVICES

GFDL Onboarding Committee *September 2024 – Present*

- Participated Postdoc Journey Mapping activities to identify difficulties of new postdocs and employees and provide suggestions to improve the onboarding process.
- Hosted “Graduate School Application Workshop” and “Good Presentation and Data Visualization Workshop” for 2025 summer interns

CLaSP-Fate High School Outreach *October 2023*

- Volunteered to host ~30 10th graders from the Jalen Rose Leadership Academy in the department for exciting hands-on activities and lab tours.
- Planned the event and proposed physics demonstrations.
- Conducted physical demonstrations and explained the physics to students.

Daily Email Group for International Students *October 2019 – October 2020*

- Created and organized the group where members write emails to other group members at any frequency they like to share their life, experiences, and stories.
- Named to be the English Language Institute (ELI) Student of the Month in December 2019. <https://lsa.umich.edu/eli/news-events/all-news/dec19studentofthemonth.html>

SKILLS

- Expertise in programming languages for scientific research, app development, and website building: Python, Fortran, C/C++, Visual Basic, NCL, JavaScript, Node.JS
- Expertise in modern software engineering technology and home server setup including version control tools (e.g., Git) and virtualization tools (e.g., PVE, Docker)
- Expertise in radiative transfer theory, satellite data analyses, and climate modeling (including CESM, E3SM, GFDL AM4)
- Independent scientific research and also a dedicated team member when necessary
- Experience in video production (Adobe Premiere) and audio editing (Adobe Audition)